UNIT 2: SHAPES AND REGIONS

SESSION 5: ACTIVE CONTOURS

IMPLEMENT AND EVALUATE ACTIVE CONTOUR MODELS FOR SEGMENTING OBJECTS IN BINARY AND GRAYSCALE IMAGES.

Snake Path Finder

An active contour (snake) starts near an object and moves toward the boundary by minimizing energy.

Observe the grid below:

0 0 0 0 0 0

0 0 1 1 1 0

0 0 1 1 1 0

0 0 1 1 1 0

0 0 0 0 0 0

The snake starts at position (1,1).

Task

1. Mark the direction the snake should move to reach the object boundary.
2. Write the coordinates of the first three positions the snake will move to.

Step 1: (1,2)

Step 2: (1,3)

Step 3: (1,4)

Energy Minimization

Active contour models use three types of energy.

Unscramble the words to identify them:

1. NITRAELN YRENEG → Internal Energy
2. AXRETENL YRENEG → External Energy
3. EGDYEAB YRENEG → Edge-Based Energy